SOCIOL 723: Social Statistics II

Week 10, Panel Data and Difference-in-Differences

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Today

Panel Data and Unobserved Heterogeneity

Fixed or Random Intercepts, Without Time

Difference-in-Differences

The Trouble with Two-Way Fixed Effects

Beyond a Binary, Absorbing Treatment

Applications in Sociology

Practice

★ = essential, you must understand this ★ = for understanding, not examined

A Third Source of Variation

So far: adjust for observables (Week 6), find an instrument (Week 7), find a threshold (Week 9). Today: use variation over time within the same unit.

With $Y_{it} = \alpha_i + \lambda_t + \tau D_{it} + \epsilon_{it}$, observe a treated unit i and an untreated unit j before and after:

$$Y_{i,\text{after}} - Y_{i,\text{before}} = \tau + (\lambda_{\text{after}} - \lambda_{\text{before}}) + \Delta\epsilon_i \quad \text{treated: its stable traits } \alpha_i \text{ cancel}$$

$$Y_{j,\text{after}} - Y_{j,\text{before}} = (\lambda_{\text{after}} - \lambda_{\text{before}}) + \Delta\epsilon_j \quad \text{untreated: only the common change}$$

$$(\text{first}) - (\text{second}) = \tau + (\Delta\epsilon_i - \Delta\epsilon_j) \quad \text{the common change cancels}$$

- What is **stable** about a unit (ability, upbringing) is differenced away, measured or not: **fixed effects**.
- What **changes for everyone** (a recession, getting older) is not, but the untreated units measure it: **difference-in-differences**.
- One idea, one assumption: nothing changes differently for the treated (parallel trends).

Panel Data

The Example: Does Marriage Raise Men's Wages?

The question. Married men earn more than unmarried men. Is it marriage (specialization at home, employers' favour), or do men with high earning potential marry?

The data. NLSY men observed every year 1980–87: 545 men, 4,360 person-years (Vella and Verbeek 1998; `wooldridge::wagepan`). Y_{it}: log hourly wage of man i in year t; $D_{it} = 1$ if married that year.

men never married in the panel	162
men married in all eight years	73
men whose status changes	310
share married, 1980 → 1987	0.19 → 0.61
mean log wage, 1980 → 1987	1.39 → 1.87

(figs/w10_panel_slow.R)

Step 1: Pooled OLS

Treat the 4,360 person-years as if they were 4,360 different men:

$$Y_{it} = a + \beta D_{it} + u_{it}, \qquad \hat{\beta} = 0.220 \ (0.026)$$ married men earn about 22% more

$$\operatorname{plim} \hat{\beta} = \beta + \frac{\operatorname{Cov}(D_{it}, u_{it})}{\operatorname{Var}(D_{it})}$$ omitted variables (Week 2)

- u_{it} holds everything else that sets a man's wage: ability, ambition, schooling, family background.
- Most of it is *stable* over these eight years. If the same traits make a man more likely to marry, $\operatorname{Cov}(D_{it}, u_{it}) > 0$ and 0.220 is too big.
- We cannot measure all of them. But we see each man more than once.

Standard errors clustered by man throughout (figs/w10_panel_slow.R).

Step 2: Compare Each Man With Himself

Split the error: $u_{it} = \alpha_i + \epsilon_{it}$, with α_i the man's stable wage level. Take two years:

$$Y_{i,87} = \alpha_i + \beta D_{i,87} + \epsilon_{i,87} \qquad \text{1987}$$

$$Y_{i,80} = \alpha_i + \beta D_{i,80} + \epsilon_{i,80} \qquad \text{1980}$$

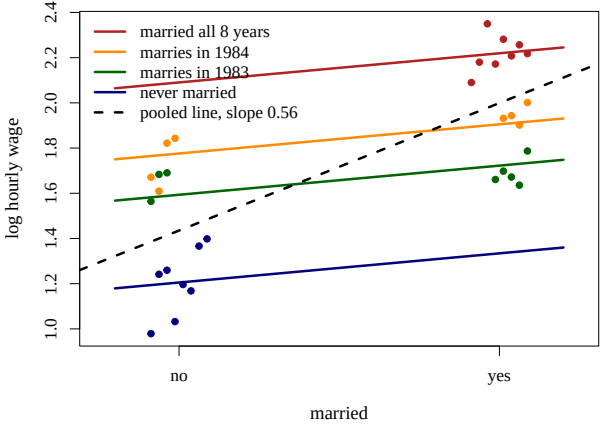
$$Y_{i,87} - Y_{i,80} = \beta\,(D_{i,87} - D_{i,80}) + (\epsilon_{i,87} - \epsilon_{i,80}) \qquad \text{subtract: } \alpha_i \text{ cancels}$$

First differences: regress each man's wage growth on his change in marital status. $\hat{\beta} = 0.445$ (0.043).

- Ability and ambition are gone, measured or not.
- But 0.445 is twice the pooled estimate. It is close to the wage growth of the 251 men who married between 1980 and 1987: 0.488. Something else changed in those seven years (Step 5).

Step 3: A Separate Intercept for Each Man

All eight years, $Y_{it} = \alpha_i + \beta D_{it} + \epsilon_{it}$: one dummy per man (545), one common slope. Four of the men:



- The pooled line (slope 0.56 for these four) mostly compares *men*: the high-wage men are the married ones.
- Each man's line (common slope 0.13) compares him only with himself.
- All 545 men: $\hat{\beta} = 0.243$ (0.022).

The Error Components Model ★

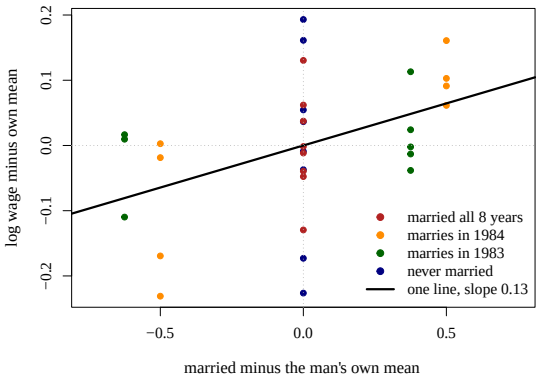
The pooled model with its error split in two (labels as in Wooldridge 2010, ch. 10):

$$Y_{it} = \beta D_{it} + \underbrace{\alpha_i}_{\text{stable: the man}} + \underbrace{\epsilon_{it}}_{\text{changes year to year}} .$$

	assumption	needed for	marriage data
FE.1	strict exogeneity: $E[\epsilon_{it} \mid D_{i1}, \ldots, D_{iT}, \alpha_i] = 0$, all t	consistency	tested below
FE.2	D_{it} varies over time within some units	identification	310 of 545 men
FE.3	$\text{Var}(\epsilon_{it} \mid D_i, \alpha_i) = \sigma^2$, no correlation across t	usual s.e.	false: cluster by man

- Nothing is assumed about α_i: it may correlate with D_{it} in any way.
- Random effects adds $E[\alpha_i \mid D_{i1}, \ldots, D_{iT}] = E[\alpha_i]$ (next section).

Step 4: Or Subtract Each Man's Mean

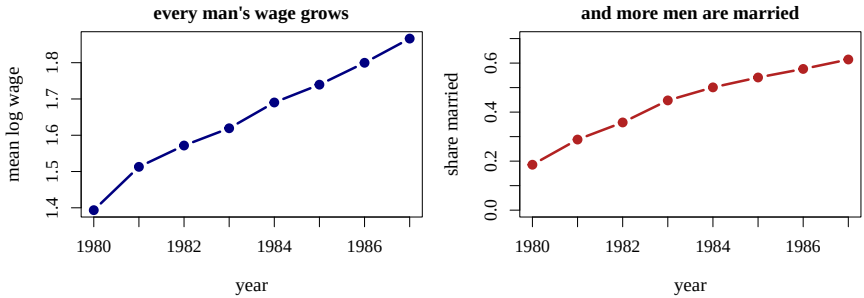


$$Y_{it} = \alpha_i + \beta D_{it} + \epsilon_{it}$$
$$\bar{Y}_i = \alpha_i + \beta \bar{D}_i + \bar{\epsilon}_i$$
$$Y_{it} - \bar{Y}_i = \beta(D_{it} - \bar{D}_i) + (\epsilon_{it} - \bar{\epsilon}_i)$$

Average each man's eight years, subtract: α_i cancels.

- OLS on the demeaned data = OLS with 545 dummies: 0.2427 both ways (FWL).
- The 235 men whose status never changes sit at $D_{it} - \bar{D}_i = 0$: they add nothing to $\hat{\beta}$.

Step 5: Why 0.24 Is Still Too Big: Time



- Every man's wage grows (experience, seniority, prices), and men marry as they age.
- A man's married years are his *later* years. Comparing them with his own earlier single years credits marriage with growth every man gets.
- α_i cannot absorb this: it is the same for all men but changes over time. Call it λ_t.

Step 5: A Control Group Measures the Growth

Back to 1980 and 1987, now with a year effect λ_t:

$$Y_{i,87} - Y_{i,80} = (\lambda_{87} - \lambda_{80}) + \beta\,(D_{i,87} - D_{i,80}) + \Delta\epsilon_i \qquad \alpha_i \text{ cancels; } \lambda \text{ does not}$$

change in married, 1980 → 1987	men	mean wage growth
married (+1)	251	0.488
no change (0)	277	0.476
left marriage (−1)	17	0.200

OLS with an intercept: $\hat{\lambda}_{87} - \hat{\lambda}_{80} = 0.451$ (0.033), $\hat{\beta} = 0.051$ (0.047).

- The intercept is the growth every man gets, measured mostly by the 277 men whose status did not change: a control group.
- $\hat{\beta}$ is the *extra* growth of men who married: difference-in-differences.

Step 6: Two-Way Fixed Effects ★

All eight years, an intercept for each man *and* each year. In a balanced panel:

$$
\begin{aligned}
Y_{it} &= \alpha_i + \lambda_t + \beta D_{it} + \epsilon_{it} && \text{the model} \\
\bar{Y}_{i\cdot} &= \alpha_i + \bar{\lambda} + \beta \bar{D}_{i\cdot} + \bar{\epsilon}_{i\cdot} && \text{average over } t\text{: man } i\text{'s mean} \\
\bar{Y}_{\cdot t} &= \bar{\alpha} + \lambda_t + \beta \bar{D}_{\cdot t} + \bar{\epsilon}_{\cdot t} && \text{average over } i\text{: year } t\text{'s mean} \\
\bar{Y} &= \bar{\alpha} + \bar{\lambda} + \beta \bar{D} + \bar{\epsilon} && \text{average over both} \\
Y_{it} - \bar{Y}_{i\cdot} - \bar{Y}_{\cdot t} + \bar{Y} &= \beta(D_{it} - \bar{D}_{i\cdot} - \bar{D}_{\cdot t} + \bar{D}) + \ddot{\epsilon}_{it} && \text{line } 1 - 2 - 3 + 4\text{: } \alpha_i \text{ and } \lambda_t \text{ cancel}
\end{aligned}
$$

- **Double demeaning**: subtract the man's mean and the year's mean, add back the grand mean (subtracted twice). Same $\hat{\beta}$ as man and year dummies: 0.0606 both ways.
- Pooled 0.220 → man effects 0.243 → man and year effects 0.061 (0.022).

Strict Exogeneity: What Demeaning Requires ★

With one regressor and unit effects only, the within estimator is

$$\hat{\beta}_{FE} - \beta = \frac{\sum_i \sum_t (D_{it} - \bar{D}_{i\cdot})(\epsilon_{it} - \bar{\epsilon}_{i\cdot})}{\sum_i \sum_t (D_{it} - \bar{D}_{i\cdot})^2} \qquad \text{substitute the demeaned model}$$

$$\bar{\epsilon}_{i\cdot} = \tfrac{1}{T}(\epsilon_{i1} + \cdots + \epsilon_{iT}) \qquad \text{contains every period's error}$$

The numerator has mean zero only if D_{is} is uncorrelated with ϵ_{it} for *every* pair s, t: past, present, and future. That is FE.1.

- $E[\epsilon_{it} \mid D_{it}] = 0$ alone is not enough.
- It fails under **feedback**: a wage shock this year (ϵ_{it}) makes marriage next year ($D_{i,t+1}$) more likely.
- If only feedback fails, FE's bias shrinks like $1/T$; FD's does not (Wooldridge 2010, ch. 10).

Testing Strict Exogeneity: Add the Lead

Under FE.1, next year's treatment cannot predict this year's outcome once α_i and this year's treatment are held fixed:

$$Y_{it} = \beta D_{it} + \delta D_{i,t+1} + X_{it}'\gamma + \alpha_i + \lambda_t + \epsilon_{it}, \qquad H_0 : \delta = 0.$$

marriage data, 1980–86 (lead needs 1987)	estimate	(se)
married this year	0.048	(0.025)
married next year	−0.0005	(0.022)

- No sign that men marry after a wage jump: $p = 0.98$.
- The s.e. (0.022) is as large as the effect: a small feedback would go unseen.
- This is the first pre-period coefficient of an event study (DiD section).

Man and year effects, union, experience^2; clustered by man (figs/w10_panel_std.R).

The Marriage Wage Premium, Four Ways

NLSY men observed every year 1980–87 (545 men, 4,360 person-years; Vella and Verbeek 1998). Log hourly wage on married, union, and year effects; standard errors clustered by man (figs/w10_panel.R).

estimator	uses	married	(se)
pooled OLS (+ education, race, experience)	all variation	0.108	(0.026)
random effects	a GLS mix	0.064	(0.017)
fixed effects	within-man change	0.047	(0.021)
first differences	year-to-year change	0.038	(0.024)

- The premium falls by more than half once each man is compared with himself: men with higher earning potential are more likely to marry.
- Hausman test of FE against RE: $\chi^2_{10} = 37.0$, $p < 0.001$.
- 310 of 545 men change marital status; the other 235 contribute nothing to the FE estimate of the marriage effect (next frames).

What Fixed Effects Averages

Binary D, unit i treated in m_i of T periods, $p_i = m_i/T$. With tildes for deviations from unit means:

$$\hat{\beta}_{FE} = \frac{\sum_i \sum_t \tilde{D}_{it}\tilde{Y}_{it}}{\sum_i \sum_t \tilde{D}_{it}^2} = \sum_i \underbrace{\frac{\sum_t \tilde{D}_{it}^2}{\sum_j \sum_t \tilde{D}_{jt}^2}}_{w_i} \underbrace{\frac{\sum_t \tilde{D}_{it}\tilde{Y}_{it}}{\sum_t \tilde{D}_{it}^2}}_{b_i}$$

multiply and divide by $\sum_t \tilde{D}_{it}^2$

$$\sum_t \tilde{D}_{it}^2 = T\, p_i(1-p_i)$$

variance of a 0/1 variable

b_i is unit i's own before–after difference. Units are weighted by how much their treatment varies, not by their number.

years married, of 8	1	2	3	4	5	6	7
weight per man, $8p(1-p)$	0.88	1.50	1.88	2.00	1.88	1.50	0.88
share of total weight	.08	.11	.19	.15	.22	.14	.12

Man effects only: $\hat{\beta}_{FE} = 0.243$ = weighted mean of 310 men's b_i (unweighted 0.255). Adding year effects: 0.061; wages rise with age for everyone, and marriage comes with age (figs/w10_panel.R).

What Fixed Effects Costs ★

- **Precision**. You are using only within-unit variation. If D_{it} barely moves within units, the standard error explodes, FWL again.
- **Measurement error is amplified**. Differencing removes signal and keeps noise, so attenuation bias gets worse.
- **Nickell bias**: adding a lagged outcome to a fixed-effects model creates its own bias (dynamic panels, below).
- Always cluster on the unit (Week 2). Serial correlation within units is the norm.

Random Trends: Each Man His Own Slope

FE allows men to differ in wage *levels*. If men on steeper wage paths marry, FE still mistakes their growth for a marriage effect.

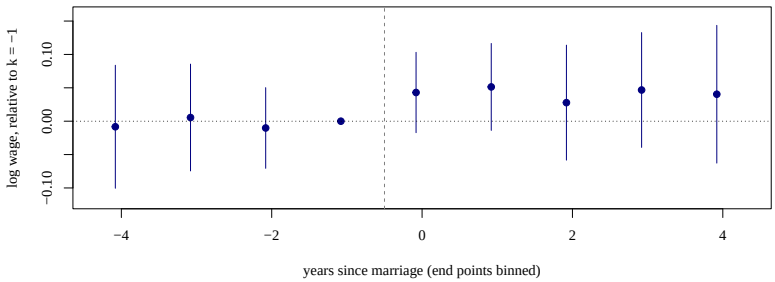
$Y_{it} = \alpha_i + \gamma_i\, x_{it} + \beta D_{it} + \epsilon_{it}$ own intercept α_i, own slope γ_i in experience x

$\ddot{Y}_{it} = \beta \ddot{D}_{it} + \ddot{\epsilon}_{it}$ double dots: residuals from man i's own regression on $(1, x_{it})$

- The **random trend model** (Wooldridge 2010, ch. 11); sociology calls it **FEIS**, fixed effects with individual slopes: detrend each unit separately, then pool (Brüderl and Ludwig 2015; Rüttenauer and Ludwig 2023). Needs at least three observations per unit.
- Identifies β from *departures* from each man's own trajectory at marriage: a jump, not a steeper path.
- Our 8-year panel: FE 0.047 (0.021), FEIS 0.055 (0.024). Eight years per man is short for estimating a slope; the NLSY79 over 34 years (next frames) tells a different story.

The Impact Function: an Event Study Around Marriage

Men who marry once in 1981–87 (239) and men never married in the panel (162); event-time dummies, man and year effects, $k = -1$ omitted (figs/w10_panel.R):



- Before marriage: −0.008, 0.006, −0.010. After: 0.04–0.05, each within noise.
- The impact function shows what one coefficient hides: anticipation, build-up, fade-out (Ludwig and Brüderl 2021). It is a DiD event study with marriage as the event (DiD section).

In the Literature: Budig and England (2001)

The question. Why do mothers earn less? NLSY79 women 1982–93: 5,287 women, 41,842 person-years; log hourly wage on number of children.

penalty per child, their Table 2	fixed effects	pooled OLS
no controls	−0.068 (0.004)	−0.081 (0.002)
+ marital status	−0.073	−0.081
+ experience, seniority, education	−0.047	−0.018
+ job characteristics	−0.037	−0.012

- About 7% per child within women, close to OLS: little selection of lower-wage women into motherhood on fixed traits.
- Experience (lost years at work) explains about a third of the FE penalty; the rest remains. Note that experience is *post*-birth, a mediator: controlling for it answers a different question.

In the Literature: Ludwig and Brüderl (2018)

The question. Is the male marriage wage premium causal, or do men on steeper careers marry? A test of specialization theory.

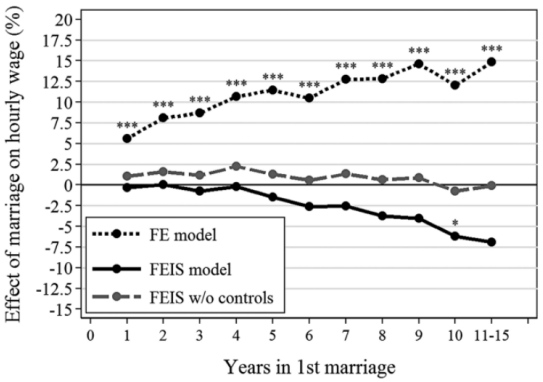
The data. NLSY79, 1979–2012: 4,287 employed men, 49,801 person-years; men observed never married at least once.

The design. Four estimators of one coefficient:

model	allows men to differ in	marriage premium
pooled OLS	nothing	17.7%
FE	wage level	8.3%
FE, group slopes	level; growth by future marital status	4.3%
FEIS	level and own wage growth	0.5% (n.s.)

Their Figure 3. Before marrying, future husbands' wages grew 5% per year of experience; never-married men's, 3%.

Ludwig and Brüderl: the Premium Over Time



Their Figure 4: effect by years since first marriage.

- FE: the premium grows to 11.4% after 5 years and 14.8% after 11–15, the pattern read as evidence of specialization.
- FEIS: about zero throughout.
- The growing "premium" is the steeper wage path the men were already on: selection on growth, not an effect of marriage.
- Generalizes: an FE impact function that keeps rising after the event is a warning sign of selection on slopes.

Fixed Effects or a Lagged Outcome?

Sociologists routinely control for last year's outcome, to absorb differences between units that are *changing* rather than fixed. It is a reasonable instinct. But combining that move with unit fixed effects creates a new bias, and it is worth knowing why before you reach for it.

- **FE and LDV embody different selection stories**: FE assumes selection on time-invariant traits; LDV assumes selection on prior outcomes. They can both be valid, but they are different assumptions and will generally give different answers.
- Angrist and Pischke (2009) suggest running both: under some conditions the two bracket the truth, so agreement is reassuring and disagreement tells you how much the selection story matters. Treat that as a heuristic, not a theorem.

Dynamic Panels: Nickell Bias

The outcome depends on its own past:

$$Y_{it} = \rho Y_{i,t-1} + X_{it}'\beta + \alpha_i + \epsilon_{it}.$$

FE.1 fails by construction: $Y_{i,t-1}$ contains $\epsilon_{i,t-1}$, which is in $\bar{\epsilon}_{i\cdot}$. Demeaned lag and demeaned error are correlated (Nickell 1981):

$$\text{plim}_{N\to\infty}\ \hat{\rho}_{FE} - \rho \ \approx\ -\frac{1+\rho}{T-1}$$

large-T approximation; exact formula in the script

simulation, $\rho = 0.5$, $N = 500$, 200 draws	$T = 4$	$T = 8$	$T = 16$
mean of $\hat{\rho}_{FE}$	-0.04	0.27	0.40
Nickell's exact fixed-T value	-0.04	0.27	0.39

Large N does not help; only large T does (figs/w10_panel_std.R).

Anderson–Hsiao: Difference, Then Instrument

First-difference to remove α_i:

$$\Delta Y_{it} = \rho\,\Delta Y_{i,t-1} + \Delta X_{it}'\beta + \Delta\epsilon_{it} \qquad \alpha_i \text{ gone}$$

$$\Delta Y_{i,t-1} = Y_{i,t-1} - Y_{i,t-2}, \quad \Delta\epsilon_{it} = \epsilon_{it} - \epsilon_{i,t-1} \qquad \text{both contain } \epsilon_{i,t-1}\text{: endogenous}$$

Instrument $\Delta Y_{i,t-1}$ with $Y_{i,t-2}$ (Anderson and Hsiao 1981), the Week 7 conditions:

- **Relevance**: $Y_{i,t-2}$ is part of $\Delta Y_{i,t-1}$.
- **Exogeneity**: $Y_{i,t-2}$ depends on errors up to $t-2$; $\Delta\epsilon_{it}$ on t and $t-1$. Holds only if ϵ_{it} is serially uncorrelated.

same simulation, $\rho = 0.5$	$T = 4$	$T = 8$	$T = 16$
mean of $\hat{\rho}_{AH}$ (sd)	0.51 (0.13)	0.51 (0.05)	0.50 (0.03)

Arellano–Bond: Use Every Valid Lag

If ϵ_{it} is serially uncorrelated, *every* lag from $t-2$ back is a valid instrument for period t's differenced equation:

$$E\big[Y_{i,t-s}\,\Delta\epsilon_{it}\big] = 0, \qquad s \geq 2,\ t = 3, \ldots, T.$$

GMM combines all these moment conditions (Arellano and Bond 1991).

- More instruments, more precision: period t has $t-2$ of them. With $T = 8$ and year effects: 29 instruments.
- **The key check**: the differenced residuals $\Delta\hat{\epsilon}_{it}$. First-order correlation is expected: $\Delta\epsilon_{it}$ and $\Delta\epsilon_{i,t-1}$ share $\epsilon_{i,t-1}$. Second-order correlation must be zero; if it is not, ϵ_{it} is serially correlated and $Y_{i,t-2}$ is invalid. Start at $Y_{i,t-3}$.
- Too many instruments overfit the first stage and pull GMM toward FE-style bias; report the count (Roodman 2009).

A Dynamic Wage Equation on the Marriage Data

log wage on its lag, married, union; year effects; robust s.e.; `fixest` (AH), `plm::pgmm` one-step (AB) (figs/w10_panel_std.R):

estimator	lagged wage $\hat{\rho}$	married	note
pooled OLS	0.60 (0.03)	0.056 (0.015)	α_i in the error: $\hat{\rho}$ too high
FE	0.06 (0.02)	0.062 (0.021)	Nickell: $\hat{\rho}$ too low
Anderson–Hsiao, Y_{t-2}	0.12 (0.03)	0.046 (0.022)	first-stage $F = 830$
Arellano–Bond, Y_{t-2} on	0.16 (0.04)	0.044 (0.023)	2nd-order test $z = 2.18$, $p = 0.03$
Arellano–Bond, Y_{t-3} on	0.68 (0.14)	0.031 (0.031)	2nd-order test $p = 0.01$
Anderson–Hsiao, Y_{t-3}	0.81 (0.27)	0.027 (0.034)	first-stage $F = 12$

- The second-order test rejects: wage shocks persist, so Y_{t-2} is not a valid instrument, and the rows using it are biased.
- Lag 3 is weak ($F = 12$) and the test still rejects: $\hat{\rho}$ jumps to 0.7–0.8 with large s.e. Persistence is not pinned down in eight years.
- The marriage coefficient, 0.03–0.06, is far less sensitive than $\hat{\rho}$.

FE vs RE

Fixed or Random Intercepts: the Question

Goal: the same model whenever members share a group (students in a school, siblings in a family, workers in a firm, the years of one life):

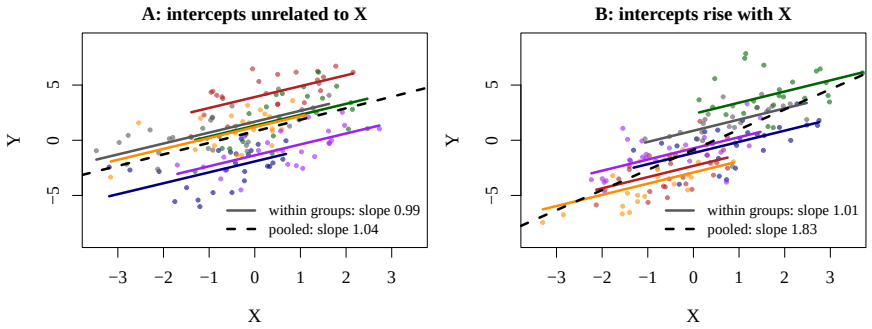
$$Y_{ij} = \alpha_j + \beta X_{ij} + \epsilon_{ij}.$$

	assumption on α_j	uses
fixed effects	none: α_j may covary with X	within groups only
random effects	$E[\alpha_j \mid X_{1j}, \ldots, X_{nj}] = E[\alpha_j]$: unrelated to X	within and between

- Both let α_j vary across groups. The names mislead: the one difference is whether α_j may covary with X (Wooldridge 2010, ch. 10).
- If it may not, random effects is more precise and estimates group-level variables.

Step 1: Two Worlds in a Picture

Six groups, $\beta = 1$; same $\mathrm{Var}(\alpha_j)$ in both worlds, only its covariance with X differs (figs/w10_fe_sim.R):



- A: high-X groups are no higher; pooled and within slopes agree (1.04 vs 0.99).
- B: high-X groups have high intercepts; the pooled slope adds that gap (1.83 vs 1.01).

Step 2: Three Estimators, One Formula

Subtract a share θ of each group's mean, then run OLS:

$$Y_{ij} - \theta\bar{Y}_j = \beta(X_{ij} - \theta\bar{X}_j) + (1-\theta)\,\alpha_j + (\epsilon_{ij} - \theta\bar{\epsilon}_j) \qquad \text{model minus } \theta\times \text{ group mean}$$

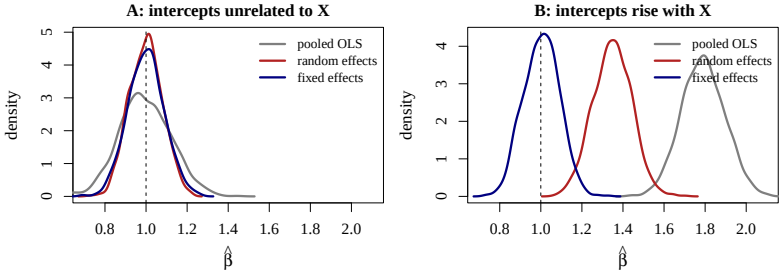
θ	estimator	share of α_j left in the error
0	pooled OLS	all
$1 - \sqrt{\sigma^2/(\sigma^2 + n\tau^2)}$	random effects (GLS)	$1-\theta$
1	fixed effects	none

$\tau^2 = \mathrm{Var}(\alpha_j)$, $\sigma^2 = \mathrm{Var}(\epsilon_{ij})$, n members per group.

- For $\theta < 1$ a share $(1-\theta)$ of α_j stays in the error. That is harmless only if α_j is unrelated to X: the random-effects assumption.
- Simulation below: $n = 4$, $\sigma = 1.5$, $\tau = 2$, so $\theta = 1 - \sqrt{2.25/(2.25 + 4 \cdot 4)} = 0.65$.
- Large groups ($n\tau^2 \gg \sigma^2$) push θ to 1: random effects approaches fixed effects.

Step 3: 1,000 Samples in Each World

100 groups of 4, $\beta = 1$:



mean (sd) of $\hat{\beta}$	pooled OLS	random effects	fixed effects
A: unrelated	0.995 (0.126)	0.999 (0.080)	1.000 (0.088)
B: rise with X	1.790 (0.105)	1.348 (0.094)	1.003 (0.087)

A: all unbiased, random effects most precise. B: random effects keeps $1 - \theta = 35\%$ of α_j and is biased; fixed effects is not.

Step 4: Testing Fixed Against Random

H_0: α_j is unrelated to X (Hausman 1978; Mundlak 1978). Fixed effects is consistent either way; random effects is consistent and efficient under H_0, inconsistent otherwise.

$$H = \frac{(\hat{\beta}_{FE} - \hat{\beta}_{RE})^2}{\mathrm{Var}(\hat{\beta}_{FE}) - \mathrm{Var}(\hat{\beta}_{RE})} \xrightarrow{d} \chi^2_1$$

Hausman: is the gap more than noise?

$$Y_{ij} = \mu + \beta X_{ij} + \gamma \bar{X}_j + (a_j + \epsilon_{ij}), \quad H_0 : \gamma = 0$$

Mundlak: $\alpha_j = \mu + \gamma \bar{X}_j + a_j$

rejection rate at 5%	A: unrelated	B: rise with X
Hausman	0.056	1.00
Mundlak, clustered s.e.	0.061	1.00

- Mundlak's $\hat{\beta}$ equals the fixed-effects $\hat{\beta}$ (FWL). Its test survives heteroskedasticity and clustering; Hausman's needs random effects to be efficient.
- On the marriage data: Hausman $\chi^2 = 37.0$; Mundlak $\hat{\gamma} = 0.094$ (next two frames).

On the Marriage Data: Random Effects

RE is OLS after subtracting a share θ of each man's mean. In a balanced panel, one θ for everyone:

$$Y_{it} - \theta \bar{Y}_{i\cdot} = (X_{it} - \theta \bar{X}_{i\cdot})'\beta + \ldots, \qquad \theta = 1 - \sqrt{\frac{\sigma_\epsilon^2}{\sigma_\epsilon^2 + T\sigma_\alpha^2}}$$

$$\hat{\theta} = 1 - \sqrt{\frac{0.123}{0.123 + 8(0.105)}} = 0.64 \qquad \text{marriage data}$$

- The man effect is 46% of the error variance; RE removes 64% of each man's mean.
- RE needs FE.1 *and* $E[\alpha_i \mid X_{i1}, \ldots, X_{iT}] = 0$: the man effect unrelated to his whole marriage history.
- RE's 0.064 lies between pooled 0.108 and FE 0.047 ("Four Ways" table): it partly uses the between-man comparison, and so can estimate education and race.

Swamy–Arora variance components, `plm` (figs/w10_panel_std.R).

On the Marriage Data: Mundlak (Mundlak 1978)

Suppose the unit effect depends on the regressors only through their mean, $\alpha_i = \gamma \bar{D}_i + a_i$ with a_i unrelated to $D_{i1}, \ldots, D_{iT}$:

$$
\begin{aligned}
Y_{it} &= \beta D_{it} + \alpha_i + \epsilon_{it} && \text{the FE model} \\
&= \beta D_{it} + \gamma \bar{D}_i + (a_i + \epsilon_{it}) && \text{substitute } \alpha_i \\
&= \beta (D_{it} - \bar{D}_i) + (\beta + \gamma) \bar{D}_i + (a_i + \epsilon_{it}) && \text{regroup: within and between parts}
\end{aligned}
$$

- Pooled OLS of this equation gives exactly the FE $\hat{\beta}$ (FWL: D_{it} net of $\bar{D}_i$ is $D_{it} - \bar{D}_i$). Marriage data: 0.0467 both ways.
- γ is the between-minus-within gap: 0.094 (0.045) here. Random effects is this model with $\gamma = 0$; testing $\gamma = 0$ is a Hausman test that works with clustered errors.
- Sociology's **hybrid model** (Allison 2009) is the last line plus time-invariant covariates, which FE cannot estimate; their coefficients are not causal merely because the model is FE-consistent.

Fixed or Random: How to Choose

if the question is	use
the effect of a member-level variable, and the group effect may be confounded	fixed effects (or Mundlak: same $\hat{\beta}$)
the effect of a group-level variable (school sector)	Mundlak with random effects: γ protects β, but the group-level coefficient still needs a_j unrelated to it
how much variation lies between groups (Week 4's ICC)	random effects

- Not rejecting does not make fixed effects wrong: it stays consistent either way.
- Fixed effects is not automatically causal: it removes only what group members share.

DID

The Canonical 2×2 ★

Two groups, two periods. One group is treated in period 2.

	Before	After	Difference
Treated	$E[Y \mid g{=}1, t{=}1]$	$E[Y \mid g{=}1, t{=}2]$	Δ_1
Control	$E[Y \mid g{=}0, t{=}1]$	$E[Y \mid g{=}0, t{=}2]$	Δ_0
			$\tau = \Delta_1 - \Delta_0$

- The first difference removes anything fixed about each group.
- The second difference removes anything common to both groups over time.
- Equivalent regression: $Y_{it} = \alpha + \gamma\,\text{treated}_i + \delta\,\text{post}_t + \tau\,(\text{treated}_i \times \text{post}_t) + \epsilon_{it}$.

The 2×2 in the Literature: Card and Krueger (1994)

New Jersey raised its minimum wage from \$4.25 to \$5.05 in April 1992; Pennsylvania did not. 410 fast-food restaurants surveyed in February and November 1992.

FTE employment per store (se), their Table 3	PA (79)	NJ (331)	NJ − PA
February (before)	23.33 (1.35)	20.44 (0.51)	−2.89 (1.44)
November (after)	21.17 (0.94)	21.03 (0.52)	−0.14 (1.07)
change	−2.16 (1.25)	0.59 (0.54)	2.76 (1.36)

- $2.76 = 0.59 - (-2.16)$: Pennsylvania's decline stands in for what New Jersey would have done without the law.
- Parallel trends here is one claim about one region in one season. Their within-NJ check: high-wage NJ stores, not bound by the law, changed like PA (−2.16 vs. −2.28, balanced sample).

The 2×2 in Regression Form

Write $g_i \in \{0, 1\}$ for group and $t \in \{1, 2\}$ for period:

$$Y_{it} = \alpha + \gamma\, g_i + \delta\, \text{post}_t + \tau\, (g_i \times \text{post}_t) + \epsilon_{it}.$$

Read off the four cell means:

	Before	After	Difference
Treated ($g = 1$)	$\alpha + \gamma$	$\alpha + \gamma + \delta + \tau$	$\delta + \tau$
Control ($g = 0$)	α	$\alpha + \delta$	δ
			τ

- γ absorbs fixed differences in level; δ absorbs the common time trend; τ is what is left.

The 2×2 in Regression Form (cont.)

- With unit and period fixed effects instead of g_i and post_t; this is the two-way fixed-effects model, identical when there are two groups and two periods, and, as we will see, not identical otherwise.

Parallel Trends ★

The identifying assumption

$$E[Y_{i2}(0) - Y_{i1}(0) \mid g_i = 1] = E[Y_{i2}(0) - Y_{i1}(0) \mid g_i = 0].$$

Absent treatment, the two groups' average outcomes would have moved in parallel.

- Levels may differ arbitrarily. Only the *trends* in the untreated potential outcome must agree.
- Untestable: it concerns a counterfactual trend. Pre-treatment parallelism is suggestive evidence, not proof.
- **Not scale-free**: parallel trends in levels and parallel trends in logs are different assumptions, and both cannot generally hold. The next slide shows why this is a real choice and not a technicality.

Parallel Trends ★ (cont.)

- Needs **no anticipation**: units must not change behaviour before treatment in expectation of it.

What the Double Difference Identifies, Exactly ★

Decompose the treated group's observed change. Its period-2 outcome is $Y_{i2}(1)$; add and subtract the counterfactual $Y_{i2}(0)$:

$$\Delta_1 = E[Y_{i2}(1) - Y_{i1}(0) \mid g_i{=}1] = \underbrace{E[Y_{i2}(1) - Y_{i2}(0) \mid g_i{=}1]}_{\text{ATT}} + \underbrace{E[Y_{i2}(0) - Y_{i1}(0) \mid g_i{=}1]}_{\text{treated group's } \textit{untreated} \text{ trend}}.$$

The control group supplies an estimate of that second, unobservable term: $\Delta_0 = E[Y_{i2}(0) - Y_{i1}(0) \mid g_i{=}0]$. Subtracting,

$$\boxed{\Delta_1 - \Delta_0 \;=\; \text{ATT} \;+\; \underbrace{E[\Delta Y(0) \mid g{=}1] - E[\Delta Y(0) \mid g{=}0]}_{\text{differential trend; } = 0 \text{ under parallel trends}}}$$

- The estimand is the ATT, not the ATE: we borrowed a counterfactual trend *for the treated group*. Nothing in the design identifies what treatment would have done to the controls.

What the Double Difference Identifies, Exactly ★ (cont.)

- The bias term *is* the parallel-trends violation, exactly. Every diagnostic in this lecture (pre-trends, placebos, Rambachan–Roth bounds) is aimed at that one object.
- Note also what was *not* assumed: no restriction on the level difference γ, and no randomization anywhere.

Parallel Trends Is Not Scale-Free ★

Suppose untreated outcomes would have grown by 10% in both groups, from a base of 20 in the treated group and 40 in the control group.

	Before	After	Change
Treated	20	22	+2
Control	40	44	+4

- Parallel trends holds in logs (both +10%) but fails in levels (+2 versus +4). A DID in levels would report a spurious −2.
- The two assumptions are mutually exclusive unless base levels are equal. There is no neutral choice, and the transformation is part of the identifying assumption, not a presentational detail.

Parallel Trends Is Not Scale-Free ★ (cont.)

- Choose on substantive grounds, is the intervention plausibly additive or multiplicative?, and *before* looking at the results.
- One diagnostic: if pre-treatment levels are far apart, be especially careful. If they are similar, the choice matters less.

When Does the Scale Not Matter? (Roth and Sant'Anna 2023)

Parallel trends in levels, in logs, in ranks: when can all hold at once?

Result

Parallel trends holds for *every* monotone transformation of Y if and only if the whole *distribution* of $Y(0)$ satisfies a parallel-trends condition. Essentially only two cases deliver that: treatment is as good as random, or $Y(0)$'s distribution does not change over time for the groups that differ.

- In any real DiD the scale is therefore part of the assumption. Choose it before seeing results, on substantive grounds (previous frame).
- The distributional condition is testable in the pre-period: compare the groups' outcome distributions, not only means.
- For effects on whole distributions under weaker assumptions: Callaway and Li (2019) (later frame).

Parallel Trends Given Covariates: Three Estimators

Assume parallel trends only within levels of pre-treatment X; two periods, $\Delta Y_i = Y_{i2} - Y_{i1}$, $e(x) = \Pr(D_i = 1 \mid X_i = x)$, $m_{0,\Delta}(x) = E[\Delta Y_i \mid X_i = x, D_i = 0]$:

$$\text{regression:} \quad \text{ATT} = E[\Delta Y_i \mid D_i = 1] - E[m_{0,\Delta}(X_i) \mid D_i = 1]$$

$$\text{weighting:} \quad \text{ATT} = E\Big[\frac{D_i - e(X_i)}{\Pr(D=1)\,(1 - e(X_i))}\,\Delta Y_i\Big]$$

$$\text{doubly robust:} \quad \text{ATT} = E\big[(w_1(D_i) - w_0(D_i, X_i))\,(\Delta Y_i - m_{0,\Delta}(X_i))\big]$$

with $w_1 = D/E[D]$ and $w_0 \propto e(X)(1-D)/(1-e(X))$, normalized to mean one (Heckman, Ichimura, and Todd 1997; Abadie 2005; Sant'Anna and Zhao 2020).

- This is Week 6 applied to ΔY: matching and weighting on the *change*, not the level.
- The DR version is consistent if either e or $m_{0,\Delta}$ is right (`DRDID`); it is what `did` uses inside each $ATT(g,t)$.

Why Not Just Add Covariates to TWFE?

$Y_{it} = \alpha_i + \lambda_t + \beta D_{it} + X_{it}'\delta + \epsilon_{it}$ is the default. Even with two periods it fails when (Caetano and Callaway 2024):

1. X_{it} is affected by treatment (a "bad control", Week 5);
2. trends depend on the *level* of X, not only its change: the within transformation keeps only changes;
3. trends depend on time-invariant X (region, education), which the unit effects absorb;
4. effects vary with X, so the regression weights the subgroups in its own way (Week 6's regression weights again);
5. the linear form is wrong.

Fix: use pre-treatment X (or X at a baseline period) in one of the three estimators on the previous frame; never post-treatment X.

Inference: Placebo Laws on Real Data

Bertrand, Duflo, and Mullainathan (2004): give fake laws to random states in CPS wage data, 1979–99. Ordinary standard errors on the micro data reject "no effect" 67.5% of the time at 5%; 44% even after aggregating to a state-year panel.

Same experiment on our state panel (female suicides, 51 states, 1964–96): 25 random states get a fake law from a random year, 1,000 times (figs/w10_placebo.R).

standard error	share of fake laws "significant" at 5%
assumes independent state-years	0.477
clustered by state	0.049
clustered by state, only 3 treated states	0.139

- Residuals are serially correlated (lag-1 correlation 0.36): 33 years of one state are far fewer than 33 independent observations.
- With few *treated* clusters, clustering also fails; use randomization inference or methods built for it (Conley and Taber 2011; MacKinnon and Webb 2018).

Event Studies ★

Generalize to many periods by estimating an effect for each period relative to treatment:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \tau_k \cdot \mathbb{1}\{t - g_i = k\} + \epsilon_{it},$$

where g_i is unit i's treatment date and $k = -1$ is the omitted reference.

- **Pre-period coefficients** ($k < 0$) are the parallel-trends check. Flat and near zero is what you want to see.
- **Post-period coefficients** trace out the dynamics: does the effect grow, fade, or appear only after a lag?
- Caution: a flat pre-trend is weak evidence. The test often has low power, and conditioning your analysis on passing it distorts inference (Roth 2022). Report the power of your pre-trend test.

Event Studies: Specification Details That Matter

That specification looks innocent. Three choices inside it change the picture, and each must be reported.

- **Normalization**: one lead must be omitted (here $k = -1$). Every τ_k is relative to it, so the picture shifts if you change the reference period. Say which you used.
- **Binning the endpoints**: with unbalanced exposure, far leads and lags are estimated off few units. Bin them ($k \leq -5$, $k \geq 5$) or restrict to a balanced window, and say which.
- **Never-treated units** anchor the time effects λ_t. If there are none, the λ_t are identified only off variation in timing, and the contamination problem below is unavoidable.

Pre-Trend Tests Have Low Power (Roth 2022)

The practice: estimate the leads, see that they are insignificant, proceed as if parallel trends holds.

The audit: 12 event-study papers in leading economics journals (2014–18) with replication data. Roth asks what happens under a linear violation of parallel trends that the usual pre-test would catch only half the time.

- The bias such a violation produces is often as large as the estimated effect, sometimes larger.
- In the worst case a nominal 95% interval covers the truth 24% of the time.
- **Conditioning on passing makes it worse**: among draws that pass the pre-test, the bias is usually larger, sometimes twice as large, because passing selects draws whose noise happened to offset the trend.

Practical: report the power of the pre-test against violations you care about (`pretrends`), and replace "test and proceed" with sensitivity analysis (Rambachan and Roth 2023, below).

Base Periods: Why Event-Study Plots Disagree (Roth 2026)

No effect at all; untreated outcomes of the treated group drift by γ per period relative to controls; treatment starts at $r = 0$.

universal base ($r = -1$):	$\hat{\beta}_r = \gamma\,(r+1)$ for every r	as TWFE: one straight line
varying base, $r < 0$:	$\hat{\beta}_r = \gamma$	compares r with $r-1$
varying base, $r \geq 0$:	$\hat{\beta}_r = \gamma\,(r+1)$	compares r with -1

- Callaway–Sant'Anna's default ("varying") draws a flat pre-period and a rising post-period: a kink at zero that looks like an effect. The imputation estimator's default draws a jump.
- The pre-period points answer different questions under the two bases. Do not read a varying-base plot with TWFE eyes, and do not feed it to `HonestDiD`.
- For plots: `base_period = "universal"` in `did`. All event studies in this lecture use it.

Drawing an Event Study (Freyaldenhoven, Hansen, Pérez Pérez, and Shapiro 2025)

- **Normalize** $\hat{\beta}_{-1} = 0$ and label that point with the outcome's mean, so readers can judge magnitudes.
- **Two kinds of bands**: pointwise 95% intervals, and a sup-t band that covers the whole path at once. A path shape ("no pre-trend at all leads") is a joint claim; only the uniform band tests it.
- **Report two p-values** in the legend: no pre-trends (leads jointly zero) and effects levelling off (last lags equal).
- **Show the least wiggly confound** consistent with the leads: if a smooth trend through the pre-period would explain the post-period, the plot should show it.
- **Bin or trim the end points** and say which.

Freyaldenhoven, Hansen, and Shapiro (2019): a covariate affected by the confound but not by the policy can identify the effect even with pre-trends (instrument the confound with leads of the policy).

In the Literature: Kleven, Landais, and Søgaard (2019)

The question. How much of the gender gap in earnings is caused by children? Danish registers 1980–2013; parents with a first birth 1985–2003, followed from 5 years before to 10 years after.

The design: an event study around the first birth, separately for mothers and fathers, with age and year dummies:

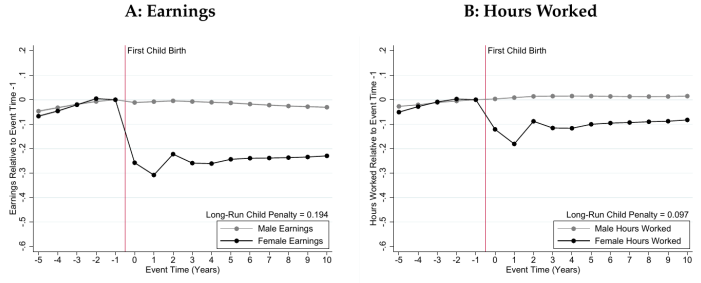
$$Y_{ist}^g = \sum_{j \neq -1} \alpha_j^g \mathbb{1}\{j = t\} + \sum_k \beta_k^g \mathbb{1}\{k = \text{age}_{is}\} + \sum_y \gamma_y^g \mathbb{1}\{y = s\} + \nu_{ist}^g$$

$$P_t = (\hat{\alpha}_t^m - \hat{\alpha}_t^w)/E[\tilde{Y}_{ist}^w \mid t]$$

with t event time, s year, and $\tilde{Y}$ the prediction from age and year alone. P_t is the **child penalty**: the gap that opens, as a share of women's counterfactual earnings.

- No control group of non-parents: identification is smoothness, the same argument as RD in time, with fathers as a check.

Kleven, Landais, and Søgaard: Results



Their Figure 1, panels A–B (accepted manuscript, 2018).

- Mothers' earnings fall almost 30% at birth and settle about 20% below the counterfactual; fathers' do not move. Long-run penalty: earnings 0.194, hours 0.097, participation 0.130, wage rate 0.091.
- The share of gender earnings inequality due to children rose from about 40% (1980) to about 80% (2013).

Staggered

The Setting Everyone Uses ★

Units adopt the policy at *different times*: states pass a law in 1995, 2001, 2008. The standard specification is

$$Y_{it} = \alpha_i + \lambda_t + \beta\, D_{it} + \epsilon_{it}, \qquad D_{it} = \mathbb{1}\{t \geq g_i\}.$$

For thirty years this was assumed to estimate "the" treatment effect. It does when the effect is constant across units and over time since treatment , but with heterogeneous or dynamic effects it need not estimate any sensible average at all.

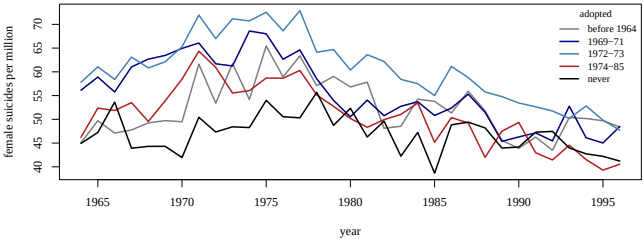
Goodman-Bacon (2021)

The TWFE estimate is a weighted average of *all possible* 2×2 DID comparisons in the data, including comparisons that use already-treated units as controls.

Running Example: Unilateral Divorce and Female Suicide

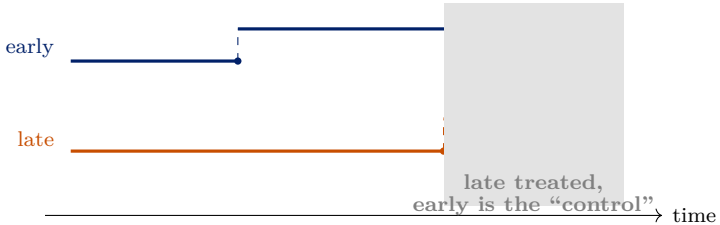
Stevenson and Wolfers (2006): did letting either spouse end a marriage reduce women's suicide? States adopted unilateral divorce in different years.

- 51 states, 1964–96; outcome: female suicides per million women (mean 51 in 1964).
- 9 states adopted before 1964 (always treated), 37 adopted 1969–85, 5 never.



Mean rate by adoption cohort. Their event study (percent change): −9.7% (2.3) averaged over 20 years, −18.7% (3.2) after 19+ years; nothing for men. Data: `bacondecomp::divorce`, crude rates (figs/w10_divorce.R).

The Forbidden Comparison ★



- In the shaded window, TWFE uses the *early-treated* unit as a control for the late-treated one.
- If the early unit's effect is still evolving, that evolution is subtracted from the late unit's effect, and enters with a negative weight.
- With dynamic effects, TWFE can be the wrong *sign* even when every unit-level effect is positive.

The Forbidden Comparison, in Algebra ★

Take the late-versus-early 2×2 from the shaded window: late switches from untreated to treated, early is *treated throughout*. Suppose parallel trends holds exactly and untreated outcomes are flat, so every change in Y is a treatment effect. Then

$$\hat{\beta}^{2\times2}_{\text{late vs. early}} = \underbrace{\text{ATT}_{\text{late}}(\text{post})}_{\text{what we want}} - \underbrace{\Big[\text{ATT}_{\text{early}}(\text{post}) - \text{ATT}_{\text{early}}(\text{pre})\Big]}_{\text{the } \textit{growth} \text{ in early's effect over the window}}.$$

With numbers; no noise, no trends, every true effect positive:

	$t=3$	$t=4$	Change
Late (treated at t=4; effect 1)	0	1	+1
Early (treated at t=2; effects 1, 2, 3)	2	3	+1
			$\hat{\beta}^{2\times2} = 0$

The Forbidden Comparison, in Algebra ★ (cont.)

- Late's true effect is $+1$; this comparison returns 0. If early's effect grew by 2 per period instead, it would return -1.
- If effects were constant $(1, 1, 1)$, the bracketed term would vanish and the comparison would be harmless, which is exactly why the problem stayed invisible for thirty years of constant-effect thinking.

The Goodman-Bacon Decomposition ★

With staggered timing, the TWFE estimate is a weighted average of every possible 2×2 comparison in the data:

$$\hat{\beta}^{\text{TWFE}} = \sum_{\text{pairs}} s_{ab}\, \hat{\beta}^{2\times 2}_{ab}, \qquad \sum s_{ab} = 1,$$

where the weights s_{ab} depend on group sizes and on the *variance of treatment* within each pair, largest for pairs whose treatment switches mid-window. (The next frame writes them out.)

Three kinds of comparison arise:

1. Treated versus **never-treated**, clean.
2. **Earlier** treated versus **later** treated, before the later group switches, clean.
3. **Later** treated versus **earlier** treated, after the earlier group switches, the forbidden comparison.

Type 3 uses already-treated units as controls. If their effects are still evolving, that evolution is subtracted from the late group's effect.

The Decomposition on the Divorce Data

TWFE: $\hat{\beta} = -3.05$ (2.32). All 156 2×2 comparisons, grouped (figs/w10_divorce.R; Goodman-Bacon's own numbers, age-adjusted rates, 49 states, in grey):

comparison	weight	average 2×2	weight	average
timing states vs. always treated (pre-1964)	0.41	−6.92	0.38	−7.04
treated vs. never treated	0.23	−5.21	0.24	−5.33
earlier treated vs. later (not yet) treated	0.11	1.25	0.11	−0.19
later treated vs. earlier (already) treated	0.25	3.36	0.26	3.51
TWFE = weighted sum		−3.05		−3.08

- A quarter of the weight sits on comparisons that use already-treated states as controls; their average has the *opposite sign*.
- 41% comes from always-treated states as controls: their whole period is post-treatment, so they are clean only if their effects are constant.

The Goodman-Bacon Weights, Explicitly ★

Let n_k, n_l, n_U be the groups' sample shares and $\bar{D}_j$ the fraction of the panel that group j spends treated (k treated earlier than l). Up to a common normalization, the weight on each 2×2 is the product of the pair's group sizes and the *variance of the treatment dummy within that pair's window*:

$$s_{kU} \propto n_k\, n_U\, \bar{D}_k(1-\bar{D}_k) \qquad \text{treated vs. never-treated}$$

$$s^k_{kl} \propto n_k\, n_l\, (\bar{D}_k-\bar{D}_l)(1-\bar{D}_k) \qquad \text{early vs. late, before late switches}$$

$$s^l_{kl} \propto n_k\, n_l\, \bar{D}_l(\bar{D}_k-\bar{D}_l) \qquad \text{late vs. early, after early switches}$$

- Every weight has the $\bar{D}(1-\bar{D})$ shape of a Bernoulli variance, so groups treated near the middle of the panel dominate: the weights come from OLS mechanics, not from group sizes alone and not from any estimand you chose. Compare Week 1's variance-weighting result: same phenomenon, new costume.

The Goodman-Bacon Weights, Explicitly ★ (cont.)

- The forbidden weight s_{kl}^{l} is not negligible in general: it grows with $\bar{D}_k$, i.e. with how long the early group has already been treated when the late group switches.
- All of this is computable from the data alone: `bacondecomp::bacon()` lists every 2×2, its estimate, and its weight. We decompose a real TWFE estimate in lab.

Where the Negative Weights Come From ★

FWL (Week 1) reduces TWFE to a bivariate regression on the *double-demeaned* treatment. With $\bar{D}_i$, $\bar{D}_t$, $\bar{D}$ the unit, period, and grand means of D_{it}:

$$\hat{\beta}^{\text{TWFE}} = \frac{\sum_{it} \tilde{D}_{it}\, Y_{it}}{\sum_{it} \tilde{D}_{it}^2}, \qquad \tilde{D}_{it} = D_{it} - \bar{D}_i - \bar{D}_t + \bar{D}.$$

Substitute $Y_{it} = \alpha_i + \lambda_t + \tau_{it} D_{it} + \epsilon_{it}$: the fixed effects are annihilated because $\tilde{D}$ has zero unit and period sums, and $\sum_{it} \tilde{D}_{it}^2 = \sum_{(i,t):D_{it}=1} \tilde{D}_{it}$, so

$$\text{plim}\ \hat{\beta}^{\text{TWFE}} = \sum_{(i,t):\, D_{it}=1} w_{it}\, \tau_{it}, \qquad w_{it} = \frac{\tilde{D}_{it}}{\sum_{(i,t):\, D_{it}=1} \tilde{D}_{it}}, \qquad \sum w_{it} = 1.$$

- For a *treated* cell, $\tilde{D}_{it} = 1 - \bar{D}_i - \bar{D}_t + \bar{D}$, which is negative exactly when $\bar{D}_i + \bar{D}_t > 1 + \bar{D}$: a unit treated for most of the panel, observed in a period when most of the sample is treated. That is the early adopter, late in the panel.

Where the Negative Weights Come From ★ (cont.)

- Intuition: OLS rewards cells whose treatment status is *surprising* given their unit and period averages. An early adopter's late-period treatment is completely unsurprising, so its effect is priced at zero, or below zero.
- The weights are a mechanical fact about OLS and are there even under constant effects, harmlessly. Heterogeneous τ_{it} is what turns strange weights into a wrong answer.

Negative Weights: Consequences and Diagnostics ★

de Chaisemartin and D'Haultfœuille (2020) generalize the previous frame's calculation, allowing covariates and non-binary treatments, and show that negative weights are the rule in staggered designs, not a corner case.

- Consequence: TWFE can have the wrong sign even when every $\tau_{it} > 0$. This is not a small-sample problem; it survives $n \to \infty$, because the weights, not the noise, are at fault.
- By the formula, the negative weights fall on units treated *early* and observed over a *long* post-period, exactly the units carrying the most information about long-run effects. TWFE quietly discounts, and then inverts, the long run.

Negative Weights: Consequences and Diagnostics ★ (cont.)

- **Diagnostic**: compute the weights (`TwoWayFEWeights::twowayfeweights`) and report how many are negative and what share of the total mass they carry. Small negative mass is reassuring about *this particular* problem; it says nothing about parallel trends or about dynamic-effect contamination, so it is one diagnostic, not a validity test.
- Sun and Abraham (2021) show the same contamination in event-study coefficients: with staggered timing, the TWFE $\hat{\tau}_k$ at lag k is a weighted combination of *every* cohort's effects at *every* lag, with weights that need not be positive or sum to one lag by lag. So apparent pre-trends can be artefacts of post-treatment dynamics, and a clean pre-trend plot from raw TWFE proves nothing.

TWFE is fine when treatment timing is common, or when effects are genuinely constant. It is unreliable exactly where sociologists use it most.

Negative Weights on the Divorce Data

The formula two frames back, computed cell by cell: $w_{it} = \tilde{D}_{it} / \sum_{D_{it}=1} \tilde{D}_{it}$ with $\tilde{D}$ the double-demeaned treatment (figs/w10_divorce.R).

treated state-years	1,164
with a negative weight	267
sum of the negative weights	−0.38
states carrying them	adopted before 1964, 1969, 1970
years	1974–1996

- Exactly where the formula said: early adopters, late in the panel. Their long-run effects, which Stevenson and Wolfers find are the largest, enter TWFE with the wrong sign.
- Positive weights sum to 1.38, so TWFE is not a convex average of any group's effect.

Callaway–Sant'Anna: Build Clean Blocks, Then Aggregate

Define the group-time average treatment effect for cohort g at time t:

$$ATT(g,t) = E\big[Y_t(g) - Y_t(\infty) \mid G_i = g\big],$$

identified by comparing cohort g to units *not yet treated* at t (or to never-treated units), using only periods before g as the baseline. Under cohort-level parallel trends and no anticipation,

$$ATT(g,t) = E\big[Y_t - Y_{g-1} \mid G_i = g\big] - E\big[Y_t - Y_{g-1} \mid \text{not yet treated at } t\big] :$$

a clean 2×2. The baseline $g-1$ predates cohort g's treatment, the comparison group has not started treatment by t, and no already-treated unit is ever used as a control.

- Under *conditional* parallel trends, each block can be estimated by IPW, outcome regression, or a doubly robust estimator, Week 6, reappearing inside a DID.

Callaway–Sant'Anna: Build Clean Blocks, Then Aggregate (cont.)

- **Then choose an aggregation**, and the choice is an estimand choice:
  - *simple*: average over all treated (g, t);
  - *dynamic*: average by $t - g$, giving an event-study path;
  - *group*: average within cohort, showing who benefited;
  - *calendar*: average within period, showing when.
- Sun–Abraham reaches the same place through an interaction-weighted regression; Borusyak–Jaravel–Spiess by imputing $Y(0)$ from untreated cells.

Imputation, Step by Step (Borusyak, Jaravel, and Spiess 2024)

Use only untreated cells to learn the untreated model, then fill in:

1. $Y_{it} = \alpha_i + \lambda_t + \epsilon_{it}$ on cells with $D_{it} = 0$ — never-treated, and pre-periods of the treated
2. $\hat{Y}_{it}(0) = \hat{\alpha}_i + \hat{\lambda}_t$ for cells with $D_{it} = 1$ — impute the missing potential outcome
3. $\hat{\tau}_{it} = Y_{it} - \hat{Y}_{it}(0), \quad \hat{\tau} = \sum_{D_{it}=1} v_{it}\,\hat{\tau}_{it}$ — average with weights you choose

- No treated cell is ever a control; any aggregation (v_{it}) is allowed, so the estimand is chosen, not inherited from OLS.
- **Efficient** when errors are homoskedastic and serially uncorrelated: it uses all pre-periods, not just $g - 1$.
- **Pre-trend tests are separate**: run on untreated cells only, so testing does not change the estimate.
- Gardner (2022)'s two-stage DiD is the same idea as two regressions (`did2s`).

Extended TWFE: the Regression Was Not the Problem

(Wooldridge 2025)

TWFE failed because it forced one β on all treated cells. Give every cohort–period its own coefficient:

$$Y_{it} = \alpha_i + \lambda_t + \sum_g \sum_{t \geq g} \tau_{gt} \, \mathbb{1}\{G_i = g\} \, \mathbb{1}\{\text{period} = t\} + \epsilon_{it}.$$

- The unit and period effects are then fitted on untreated cells only (every treated cell has its own dummy), so $\hat{\tau}_{gt}$ equals the imputation estimate. Divorce data: both give -4.85.
- Adding the pre-period interactions too, against never-treated states, reproduces Sun–Abraham and Callaway–Sant'Anna with a universal base: all three give -10.56.
- Covariates enter as cohort-by-period interactions (the "two-way Mundlak" regression). Nonlinear versions (logit, Poisson) follow (`etwfe`).

Stacked DiD and Its Weights

Stacking (Cengiz, Dube, Lindner, and Zipperer 2019): for each adoption cohort, build a sub-experiment (the cohort plus clean controls, in a fixed window around adoption), stack them, and run one regression with sub-experiment-specific unit and time effects.

- Never uses already-treated controls, and is easy to explain.
- Unweighted stacking does not estimate a clean average: each sub-experiment's treated and control shares differ, so OLS weights their trends unequally (Wing, Freedman, and Hollingsworth 2024). Corrective sample weights recover a "trimmed aggregate ATT".
- Used in sociology: Walmart Supercenters and unions (Choper, Lehner, and Parolin 2026); welfare reform and children's home environments (Kalil et al. 2023), both later.

The Modern Estimators ★

All of them share one principle: never use an already-treated unit as a control. Build clean 2×2 comparisons, then aggregate with weights you choose.

Method	Idea	R
Callaway–Sant'Anna (2021)	group-time ATT(g, t), never- or not-yet-treated controls	`did`
Sun–Abraham (2021)	interaction-weighted event study	`fixest::su`
de Chaisemartin–D'Haultfœuille	clean comparisons at switch dates	`DIDmultipl`
Borusyak–Jaravel–Spiess	impute $Y(0)$ from untreated cells	`didimputat`
Stacked / cohort DID	one clean event window per cohort	manual

Callaway–Sant'Anna also allows **conditional** parallel trends: trends parallel only after adjusting for covariates, estimated by IPW, regression, or doubly robust, Week 6, reappearing.

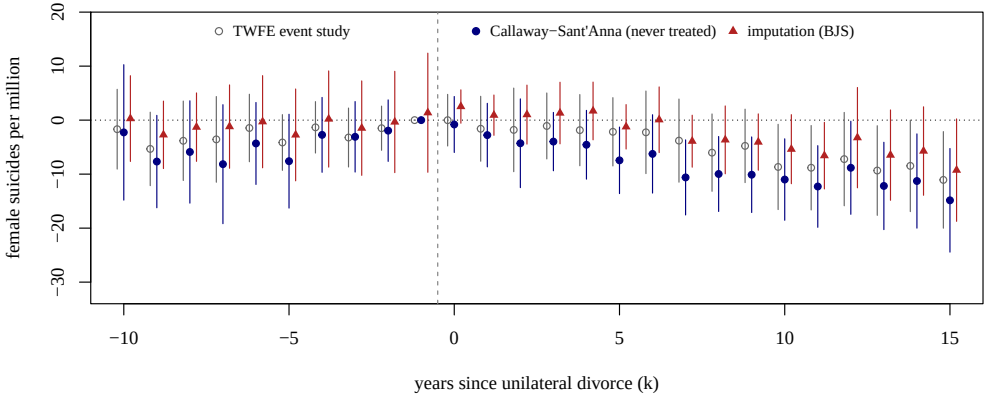
Six Estimators, One Dataset

Divorce data without the 9 always-treated states (they have no pre-period); standard errors clustered by state (figs/w10_divorce.R):

estimator	controls, baseline	ATT	(se)
TWFE, all 51 states	everything	−3.05	(2.32)
TWFE, 42 states	everything	−0.34	(2.76)
Callaway–Sant'Anna	never treated, year $g-1$	−10.56	(3.12)
Callaway–Sant'Anna	not yet treated, year $g-1$	−8.69	(3.40)
Sun–Abraham	never treated, year $g-1$	−10.56	(3.52)
imputation = extended TWFE	not yet treated, all pre-years	−4.85	(2.97)

- The robust estimators agree on the sign and say TWFE understates the effect.
- They differ in size because they assume parallel trends over different windows: from $g-1$ only, or over every pre-period. Next frame shows why that matters here.

The Event Studies, Side by Side



- Leads are not flat (Callaway–Sant'Anna at $k = -5$: -7.6): untreated trends were not parallel in every pre-year, so "which pre-years define the baseline" changes the answer.
- Effects grow with time since adoption (-14.9 at $k = 15$), the dynamic heterogeneity that breaks TWFE.

Choices Every Staggered DiD Makes

choice	options	consequence
comparison group	never treated; not yet treated	not-yet uses more data but needs no anticipation and parallel trends with future adopters
baseline	$g-1$ only; all pre-periods	all pre-periods is efficient if trends are parallel throughout, biased if not
aggregation	simple; by event time; by cohort; by calendar time	each is a different estimand; say which
anticipation	shift g back by δ periods	otherwise effects leak into the baseline
covariates	none; DR within each $ATT(g,t)$	use pre-treatment values only
inference	cluster; bootstrap; randomization	few treated cohorts need care

Efficiency when adoption timing is as good as random: Roth and Sant'Anna (2023) (`staggered`). A design-based reading of TWFE under random timing: Athey and Imbens (2022).

Honest Inference About Pre-Trends

Rather than testing pre-trends and proceeding as if they had passed, Rambachan and Roth (2023) propose bounding how badly parallel trends could fail.

- Their "relative magnitudes" restriction bounds how much the *change* in the trend violation after treatment can exceed the largest change observed before it, by a factor $\bar{M}$, and the resulting confidence sets account for sampling uncertainty in the pre-period estimates rather than treating them as known.
- Report the *breakdown value*: how large $\bar{M}$ would have to be before your conclusion reverses.
- This turns an untestable assumption into a quantitative claim, exactly the move sensitivity analysis made in Week 6.
- `HonestDiD`.

Honest Inference About Pre-Trends (cont.)

Inference reminder

Bertrand, Duflo, and Mullainathan (2004): cluster on the unit at which treatment varies. With few clusters, use CR2 or a wild cluster bootstrap (Week 2).

Sensitivity on the Divorce Data

Callaway–Sant'Anna event study, universal base; target: the average effect over $k = 0, \ldots, 10$, estimated at -6.53 (figs/w10_divorce.R, `HonestDiD`).

allowed post-period violation, relative to the largest pre-period one ($\bar{M}$)	95% interval
none (parallel trends exact)	$[-11.9, -1.1]$
$\bar{M} = 0.05$	$[-13.4, 0.8]$
$\bar{M} = 0.25$	$[-28.4, 16.4]$
$\bar{M} = 1$	$[-55.1, 55.1]$

- Breakdown below $\bar{M} = 0.05$: allowing post-period violations one-twentieth the size of the largest pre-period swing erases the sign.
- The pre-period swings are large (leads near -7.6), so relative magnitudes give little room. The estimate is honest about what these data can say.

Beyond

Doses: a Continuous Treatment

Callaway, Goodman-Bacon, and Sant'Anna (2025). Units receive dose d (minimum-wage increase in dollars, hours of a program); some get $d = 0$. Two different questions:

$$\mathrm{ATT}(d \mid d) = E[Y_t(d) - Y_t(0) \mid D = d] \qquad \text{effect of dose } d \text{ on those who got } d$$

$$\mathrm{ATT}(d \mid d) - \mathrm{ATT}(d' \mid d') = \underbrace{\mathrm{ATT}(d \mid d) - \mathrm{ATT}(d' \mid d)}_{\text{causal: more dose, same units}} + \underbrace{\mathrm{ATT}(d' \mid d) - \mathrm{ATT}(d' \mid d')}_{\text{selection: different units}}$$

- Ordinary parallel trends (each dose group vs. $d = 0$) identifies each $\mathrm{ATT}(d \mid d)$, but not how effects change with the dose.
- Comparing doses needs **strong parallel trends**: every dose group would have trended like any other under every dose. It rules out selection on gains.
- TWFE on a continuous D mixes both terms with weights that do not follow the dose distribution; its weights on level effects sum to zero (`contdid`).

Treatments That Switch On and Off

Union membership, unemployment, living with a partner: units enter and leave treatment, and past treatment may still matter.

- TWFE with D_{it} switching assumes no carryover (Imai–Kim) and constant effects, and uses switchers as each other's controls.
- de Chaisemartin and D'Haultfœuille (2026): compare units whose treatment first changes at t with units that had the *same* treatment history up to t and have not yet changed. The effect of having been exposed for ℓ periods, DID_ℓ, is an event study for non-absorbing, non-binary treatments (`DIDmultiplegtDYN`).
- Matching on treatment histories (Imai, Kim, and Wang 2023) does the same with explicit lag length L.

Cost: only units with a matching history count, so fewer comparisons and wider intervals. That is the price of not assuming carryover away.

Triple Differences

A policy hits one group (say, mothers) in treated states. Difference the state DiD for mothers against the state DiD for a group the policy should not touch (women without children):

$$\text{DDD} = \text{DiD}_{\text{mothers}} - \text{DiD}_{\text{non-mothers}}.$$

- Needs only **one** parallel-trends assumption: the bias in the two DiDs is the same. Neither DiD needs to be unbiased on its own (Olden and Møen 2022).
- With covariates or staggered timing, "the difference of two DiDs" and three-way fixed effects regressions are generally not valid; pooling not-yet-treated units can add bias even without covariates. Use estimators built for DDD (Ortiz-Villavicencio and Sant'Anna 2025; `tripledifff`).

Distributions, and a Design-Based View

Effects on the distribution (Callaway and Li 2019): with three or more panel periods, distributional parallel trends plus copula stability (the link between a unit's change in $Y(0)$ and its level stays the same over time) identify quantile effects on the treated.

Where does uncertainty come from? Standard errors here usually imagine resampling units. In policy panels we observe every state.

- Athey and Imbens (2022): treat the *adoption date* as the random element; TWFE then has a design-based reading, and the usual variance is conservative.
- Rambachan and Roth (2026): design-based uncertainty for quasi-experiments; usual standard errors can remain valid, if conservative, for a well-defined estimand.

Factor models (unit-specific responses to common shocks) relax parallel trends further: synthetic DiD, generalized synthetic control, counterfactual estimators (`fect`). Next week.

Sociology

In the Literature: Harknett, Schneider, and Irwin (2021)

The question. Does making work schedules predictable improve workers' lives? Seattle's Secure Scheduling Ordinance (2017): two weeks' notice, pay for last-minute changes, extra pay for "clopening" shifts; large retail and food-service firms.

The data. The Shift Project: surveys of hourly workers recruited by Facebook and Instagram ads. Seattle: 754 workers at baseline (spring 2017), 747 after one year, 441 after two. Comparison: workers at the *same firms* in 24 large metros with above-federal minimum wages. 17,689 responses in all.

The design. DiD with two post periods,

$$Y_{ict} = \beta_1 \,\text{Seattle}_c \times \text{Year1}_t + \beta_2 \,\text{Seattle}_c \times \text{Year2}_t + \text{city and wave effects} + X'_{ict}\delta + \epsilon_{ict},$$

linear probability models, weighted to the American Community Survey. Same firms on both sides holds employer practices fixed, as Card and Krueger held the industry fixed.

Harknett, Schneider, and Irwin: Checking the Design

Baseline levels	Similar: unpredictability scale 2.79 vs. 2.85; happy 76% vs. 75%; any material hardship 60% vs. 58%. Clopening differs (37% vs. 44%).
Pre-trends	No pre-2017 data, so no event study. Instead: in places without scheduling laws, the national scale stays near 2.8 in 2017, 2018, 2019.
Comparison group	Results hold with cities that considered but did not pass a law, with a propensity-matched sample, and (smaller) with workers just outside Seattle, where spillover is plausible.
Composition	The Seattle sample becomes slightly younger and shorter-tenured.
Sampling	Not a probability sample; more women and White workers than the workforce.

A one-treated-city DiD: inference rests on the comparison cities behaving like Seattle would have. Next week's synthetic control is built for exactly this case.

Harknett, Schneider, and Irwin: Main Results

Change in Seattle minus change in comparison cities after the law (their Figures 1–2):

outcome	DiD estimate
schedule unpredictability scale (baseline 2.8)	-0.50 (-18%)
at least two weeks' notice	$+11$ pts
last-minute changes without pay	-13 pts
on-call shifts	-7 pts $(p \leq 0.07)$
happy	$+7$ pts
good sleep quality	$+11$ pts
any material hardship	-10 pts
psychological distress	n.s.

The law changed schedules first; well-being followed. An IV extension uses the law as an instrument for predictability (first-stage $F = 17.1$).

Staggered DiD in Sociology: Two Recent Studies

Choper, Lehner, and Parolin (2026, ASR): Walmart Supercenters and unions. Restricted PSID 1983–2019; 966 first Supercenter openings across counties; stacked DiD with never-treated controls, matched workers, ±10 years.

- Union membership falls 1.4 points (se 0.8); 3.5 points (1.3) when differential linear pre-trends are removed: about 21% of the 16.1% baseline.
- The headline depends on a trend adjustment: report both, as they do.

Kalil, Corman, Dave, Schwartz-Soicher, and Reichman (2023, *Demography*): staggered state welfare reform 1992–98; NLSY79 unmarried, low-educated mothers of young children.

home environment score	TWFE	stacked (not-yet-treated controls)
emotional support	−4.95 (1.44)	−6.56 (1.77)
overall	−2.89	−5.40

Already-treated controls attenuate TWFE when effects grow, as in the divorce data.

Practice

What a Reanalysis of 49 Studies Found (Chiu, Lan, Liu, and Xu 2026)

Political scientists reran 49 published panel studies with the heterogeneity-robust estimators of this lecture.

- The new estimators usually agree with TWFE *qualitatively*, but the estimates vary a lot across estimators.
- Many studies lack evidence for parallel trends once pre-periods are examined properly.
- Many are underpowered: the robust estimators' intervals are wide.

Read with the divorce data in mind: the estimator mattered (TWFE vs. robust), and so did the window over which trends are assumed parallel. The practical guide that collects these steps: Baker, Callaway, Cunningham, Goodman-Bacon, and Sant'Anna (2026).

A DID Checklist ★

1. State the estimand: ATT for whom, over which post-treatment window?
2. Plot the raw group means over time. Before any regression.
3. Is timing staggered? If so, do not use plain TWFE. Use a modern estimator, and report the TWFE estimate only for comparison.
4. Report an event-study plot with pre-period coefficients, using Sun–Abraham or Callaway–Sant'Anna rather than raw TWFE leads.
5. Justify the scale (levels or logs) on substantive grounds.
6. Cluster on the treatment-assignment unit; report the number of clusters.
7. Report a sensitivity analysis for parallel trends.
8. Placebo: a pre-treatment outcome, or an untreated outcome that the policy should not affect.

Looking Ahead

- Fixed effects difference away time-invariant confounders, a real gain, bought with precision and with a new assumption about trends.
- The staggered-adoption literature is a case study in how a method can be standard for decades and still be wrong. Read papers published before 2021 with this in mind, including your own field's classics.
- **Next week**: what to do when there is only *one* treated unit, and "the control group" has to be constructed.
- Notice that conditional parallel trends brought Week 6 back inside a DID: once trends are only parallel given covariates, you need a weighting or outcome model again, and all of Week 6's cautions return with it.

Problem Set 4

Assigned today, due Monday November 9. Covers Weeks 7 and 9.

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